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REFERENCE GUIDE · OCTOBER 2026

Simulation Playbook

Eighteen step-by-step builds across six domains, from a single coffee counter to conveyor lines, machine bottlenecks and network load, plus thirty industry use cases for pitching.

18 guides | 6 domains | 30 use cases

For	Date	Built against
Builders, testers, sales	26 October 2026	JustCmul8 @ b1dfd7d

00 CONTENTS

Two parts: build, then pitch

Part A is a set of recipes you can follow click by click; each one ends with the numbers you should see, so it doubles as a test case. Part B lists where each domain earns its keep in industry.

Part A · Step-by-step guides

01	Coffee Shop Counter	PEOPLE & SERVICE	EASY
02	Metro Station Entry	PEOPLE & SERVICE	MEDIUM
03	Hospital OPD Patient Flow	PEOPLE & SERVICE	ADVANCED
04	Fuel Station Pumps	TRAFFIC & VEHICLES	EASY
05	Toll Plaza: FASTag vs Cash	TRAFFIC & VEHICLES	MEDIUM
06	Car Service Centre	TRAFFIC & VEHICLES	ADVANCED
07	Storm-water Retention Tank	LIQUID & MATERIAL	EASY
08	Two Tanks, One Valve	LIQUID & MATERIAL	MEDIUM
09	Batch Reactor with Quality Split	LIQUID & MATERIAL	ADVANCED
10	Single CNC Machine	MANUFACTURING	EASY
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12	Welding Bottleneck with Downtime and a Paint Conveyor	MANUFACTURING	ADVANCED
13	Conveyor Belt to Packing Station	LOGISTICS	EASY
14	Parcel Sorter with Three Chutes	LOGISTICS	MEDIUM
15	Cross-dock: Truck Waves to Outbound Docks	LOGISTICS	ADVANCED
16	Fixed Cable Latency	NETWORK & SIGNAL	EASY
17	One Message, Three Subscribers	NETWORK & SIGNAL	MEDIUM
18	Microservice Gateway Under Load	NETWORK & SIGNAL	ADVANCED

Part B · Pitch catalogue

Five use cases for each of the six domains, tagged Simple or Complex, with the business problem, what to model, the KPIs and the decision it supports.

01 HOW TO USE THIS PLAYBOOK

Read a guide in thirty seconds

Every guide has the same shape: a one-line story, the question it answers, numbered build steps with the exact panel settings, a connection map, the run length, the numbers to expect, and one experiment.

Eight rules before you build

1 · Everything is in seconds
Arrival Speed is per second (72 an hour = 0.02). Times are seconds (4 minutes = 240). Pick the run length with the unit selector next to Duration.

2 · Name every block
Rename each block to its job. Its ID (for example `node_5`) is shown at the top of its settings panel; Decision Blocks need it.

3 · Decision paths need IDs
Draw the arrows, then open the Decision Block → Outgoing Paths → **Add Path**: type the target block's ID and a chance from 0 to 1. Chances must add up to 1, or items stop there.

4 · A Waiting Line in front of every limited block
That is where waits and give-ups are measured. Patience works only when the line feeds a limited block. Line capacity and "who gets served first" are not applied by the engine today; leave them at the defaults.

5 · Time steps vs limited steps
Conveyor, Processing, Assembly and Counter steps hold any number at once: use them for belts, ovens, travel and treatment. Use a limited block when only N can work at the same time.

6 · Fold downtime into the cycle
For a machine available 90% of the time, enter cycle ÷ 0.9. The panel's breakdown fields are not applied by the main engine yet.

7 · Tanks fill, they don't drain
Each arriving batch adds its Fill Rate; a batch that does not fit is dropped. Use tanks for "when does it overflow" questions.

8 · Network guides start from templates
Channel delay and buffer size are set by the template gallery; the settings panel edits sources, decisions and exits.

Three sanity formulas

Utilization

$$\rho = \lambda \times S \div c$$

Arrival rate × time per item ÷ number of servers. ρ ≥ 1 means the line grows forever.

Throughput

$$out \approx \lambda \times T$$

When ρ < 1, everything that arrives leaves, minus what is still inside at the end.

Little's Law

$$L = \lambda \times W$$

Items inside a step = arrival rate × time spent there. Sizes belts, ovens and buffers.

Theory values move

Expected numbers are steady-state theory. A single run varies, more so near 80–90% load. When a figure is far off (for example utilization 60% where 15% is expected), it is worth a bug report; when it is within the normal spread, run more replications in Reliability Lab.

02 GLOSSARY

Same engine, domain-specific names

Each domain's palette uses its own words for the same few building blocks. Learn the eight behaviours once and every domain reads the same.

PALETTE NAME	WHERE	WHAT IT DOES AND WHAT YOU SET
Start Point	All domains	Creates arrivals: **Paced Arrivals** (pattern + Arrival Speed per second) or an **Exact Time Schedule** (at second N, release M). Several outgoing arrows take turns.
Waiting Line	People, Vehicle, Manufacturing, Logistics	Where items wait in front of a limited block; wait time and give-ups (**Will People Give Up**, patience in seconds) are measured here.
Limited Counter · Service Bay (Limited) · Machine / Robot · Loading Dock	One per domain	The same limited resource: **Number of Staff / Parallel Counters**, **Average Time to Serve** (s), **Service Speed Consistency**. Only N items at once.
Counter · Processing Block · Processing Unit · Assembly Step · Conveyor Step	One per domain	A time-only step: **Average Duration** (s) and **Speed Consistency**. Holds any number at once, so use it for belts, ovens, travel and treatment time.
Decision Block · Valve / Splitter	All domains	Sends each item down one path by chance. Set under **Outgoing Paths → Add Path**: target block ID and a chance from 0 to 1.
Storage Tank	Liquid	Accumulates **Fill Rate** units per arriving batch up to **Capacity**; batches that do not fit are dropped.
Transport Channel · Storage Rack · Broadcaster	Network (Storage Rack also in Liquid)	Delay link with a buffer · holding buffer · copy to every path. Their settings come from templates.
Exit Point	All domains	Counts completed items; **Record Final Score** includes them in the report.

A

Step-by-step guides

Eighteen builds: Easy, Medium and Advanced in each of the six domains.

GUIDE 01 PEOPLE & SERVICE EASY

Coffee Shop Counter

A café with one barista. In the morning rush a customer walks in about every 50 seconds.

QUESTION Is one barista enough, or does the line get out of hand?

#	BLOCK	NAME IT	SETTINGS	WHY
1	**Start Point**	**Customers**	Paced Arrivals · Arrival Pattern: **Natural Random Spacing** · Arrival Speed: **0.02**	0.02 per second = 72 customers an hour. Random gaps, because people don't arrive on a timetable.
2	**Waiting Line → Limited Counter**	**Order line → Barista**	Line: defaults · Number of Staff: **1** · Average Time to Serve: **40** · Consistency: **Random Service Duration**	One person, and some orders are quick while others take longer.
3	**Exit Point**	**Leaves with coffee**	Record Final Score: **On**	Customers are counted here.

```
Customers → Order line → Barista → Leaves with coffee
```

RUN Duration: **8 hrs**

CHECK THE RESULTS	THEORY	WHY
Customers arrived	≈ 576	0.02 × 28,800 s
Barista utilization	≈ 80%	0.02 × 40 ÷ 1
Average wait in line	≈ 160 s	M/M/1: ρ ÷ (μ − λ)
People waiting on average	≈ 3.2	Little's Law: 0.02 × 160

EXPERIMENT

Try: Set Number of Staff to **2**.

Expect: Utilization falls to about 40% and the average wait to about **8 seconds**. Doubling staff cut waiting by about 95%, not 50%: queues are non-linear.

> At 80% load a single run swings a lot (±40% on the wait). Use Reliability Lab → Stress with several runs for a stable average.

GUIDE 02 PEOPLE & SERVICE MEDIUM

Metro Station Entry

Evening peak at a metro station. A passenger arrives about every 5 seconds; 40% have no card and must buy a token first.

QUESTION How many token counters stop the line from growing, and how many passengers give up today?

#	BLOCK	NAME IT	SETTINGS	WHY
1	**Start Point**	**Passengers**	Arrival Pattern: **Natural Random Spacing** · Arrival Speed: **0.2**	One every 5 seconds, about 720 an hour.
2	**Decision Block**	**Has a card?**	After connecting: Outgoing Paths → Add Path → **Token line 0.4**, **Gate line 0.6**	Each passenger takes one route, by chance. Chances must add up to 1.
3	**Waiting Line**	**Token line**	Will People Give Up: **Random Patience Limits** · Average Patience Time: **300**	Real people leave after a few minutes; this measures how many are lost.
4	**Limited Counter**	**Token counter**	Number of Staff: **2** · Time to Serve: **25** · **Random Service Duration**	A clerk's time varies with each passenger.
5	**Waiting Line → Limited Counter**	**Gate line → AFC gates**	Line: defaults · Number of Staff: **4** · Time to Serve: **3** · **Fixed Constant Duration**	Four gates; a card tap takes the same time every time.
6	**Exit Point**	**Platform**	Record Final Score: **On**	Counts passengers who made it through.

```
Passengers → Has a card?
Has a card? → (0.4) Token line → Token counter → Gate line
Has a card? → (0.6) Gate line → AFC gates → Platform
```

RUN Duration: **1 hr**

CHECK THE RESULTS	THEORY	WHY
Passengers arrived	≈ 720	0.2 × 3,600 s
Token counter utilization	≈ 100%	0.08 × 25 ÷ 2 = 1.0, overloaded on purpose
Token line	keeps growing; reneges > 0	ρ ≥ 1 has no steady state
Gate utilization	≤ 15%	0.2 × 3 ÷ 4, lower as some give up

EXPERIMENT

Try: Set Token counter staff to **3**. Then save both versions in Reliability Lab → Compare.

Expect: Utilization ≈ 67%, average wait ≈ **11 seconds**, and almost nobody gives up. Compare runs each version many times, so the difference is shown with statistical confidence.

GUIDE 03 PEOPLE & SERVICE ADVANCED

Hospital OPD Patient Flow

An outpatient department sees about 36 patients an hour: registration, then a general physician (70%) or a specialist (30%), then the pharmacy.

QUESTION Where is the bottleneck, and how many general physicians keep the wait under two minutes?

#	BLOCK	NAME IT	SETTINGS	WHY
1	**Start Point**	**Patients**	**Natural Random Spacing** · Arrival Speed: **0.01**	36 an hour.
2	**Waiting Line → Limited Counter**	**Registration queue → Registration desk**	Line: defaults · Staff **2** · Time **120** · **Random Service Duration**	Two clerks, two minutes each on average.
3	**Decision Block**	**Triage**	Paths: **GP queue 0.7**, **Specialist queue 0.3**	Triage sends each patient one way.
4	**Waiting Line**	**GP queue**	**Random Patience Limits** · Patience **1800**	Some patients leave after about 30 minutes.
5	**Limited Counter**	**General physicians**	Staff **3** · Time **360** · **Random Service Duration**	Six-minute consultations.
6	**Waiting Line → Limited Counter**	**Specialist queue → Specialist**	Line: defaults · Staff **1** · Time **240** · **Random Service Duration**	
7	**Waiting Line → Limited Counter**	**Pharmacy queue → Pharmacy**	Line: defaults · Staff **2** · Time **150** · **Random Service Duration**	Both doctor paths merge here.
8	**Exit Point**	**Discharged**	Record Final Score: **On**	

```
Patients → Registration queue → Registration desk → Triage
Triage → (0.7) GP queue → General physicians → Pharmacy queue
Triage → (0.3) Specialist queue → Specialist → Pharmacy queue
Pharmacy queue → Pharmacy → Discharged
```

RUN Duration: **8 hrs**

CHECK THE RESULTS	THEORY	WHY
Patients arrived	≈ 288	0.01 × 28,800
Registration	60% · wait ≈ 68 s	M/M/2
General physicians	84% · wait ≈ 535 s	M/M/3, λ = 0.007
Specialist	72% · wait ≈ 617 s	M/M/1, λ = 0.003
Pharmacy	≈ 75% · wait ≈ 190 s	M/M/2, a little lower if GP patients leave

EXPERIMENT

Try: Compare **3 vs 4** general physicians in Reliability Lab → Compare.

Expect: With 4, GP utilization ≈ 63% and the wait drops to ≈ **79 s**. The specialist becomes the longest wait: that is the next decision.

> The clinic starts empty, so the first half hour reads light; an 8-hour run lands slightly below these steady-state figures.

GUIDE 04 TRAFFIC & VEHICLES EASY

Fuel Station Pumps

A highway fuel station sees about 72 vehicles an hour; a fill-up takes 2½ minutes on average.

QUESTION How many pumps does the station need?

#	BLOCK	NAME IT	SETTINGS	WHY
1	**Start Point**	**Vehicles**	**Natural Random Spacing** · Arrival Speed: **0.02**	72 an hour.
2	**Waiting Line → Service Bay (Limited)**	**Forecourt queue → Pumps**	Line: defaults · Staff/Parallel: **4** · Time **150** · **Random Service Duration**	Four pumps, each serving one vehicle at a time.
3	**Exit Point**	**Back on the road**	Record Final Score: **On**	

```
Vehicles → Forecourt queue → Pumps → Back on the road
```

RUN Duration: **8 hrs**

CHECK THE RESULTS	THEORY	WHY
Vehicles	≈ 576	0.02 × 28,800
Pump utilization	≈ 75%	0.02 × 150 ÷ 4
Average wait	≈ 76 s	M/M/4
Vehicles waiting	≈ 1.5	Little's Law

EXPERIMENT

Try: Drop to **3** pumps. Then add patience: Exact Maximum Wait Limit **300**.

Expect: At 3 pumps load = 100%: the queue keeps growing. With patience, drivers leave after 5 minutes and the walk-away count shows lost sales.

GUIDE 05 TRAFFIC & VEHICLES MEDIUM

Toll Plaza: FASTag vs Cash

A toll plaza handles about 900 vehicles an hour; 80% pay by FASTag, 20% in cash.

QUESTION How many cash lanes are needed, and what happens if FASTag use drops?

#	BLOCK	NAME IT	SETTINGS	WHY
1	**Start Point**	**Vehicles**	**Natural Random Spacing** · Arrival Speed: **0.25**	900 an hour.
2	**Decision Block**	**Payment type**	Paths: **FASTag queue 0.8**, **Cash queue 0.2**	
3	**Waiting Line → Service Bay (Limited)**	**FASTag queue → FASTag lanes**	Line: defaults · **2** · Time **5** · **Fixed Constant Duration**	A tag read takes the same time every time.
4	**Waiting Line → Service Bay (Limited)**	**Cash queue → Cash lanes**	Line: defaults · **2** · Time **30** · **Random Service Duration**	Cash handling varies.
5	**Processing Block**	**Exit ramp**	Average Duration **10** · **Fixed Constant Duration**	Travel time only; any number of cars can be on the ramp.
6	**Exit Point**	**Highway**	Record Final Score: **On**	

```
Vehicles → Payment type
Payment type → (0.8) FASTag queue → FASTag lanes → Exit ramp
Payment type → (0.2) Cash queue → Cash lanes → Exit ramp
Exit ramp → Highway
```

RUN Duration: **1 hr**

CHECK THE RESULTS	THEORY	WHY
Vehicles	≈ 900	0.25 × 3,600
FASTag lanes	≈ 50%	0.2 × 5 ÷ 2
Cash lanes	75% · wait ≈ 39 s	M/M/2, λ = 0.05
Cars on the ramp	≈ 2.5	Little's Law: 0.25 × 10

EXPERIMENT

Try: Change the split to **0.6 / 0.4** (FASTag use falls).

Expect: Cash load = 0.1 × 30 ÷ 2 = **150%**: the cash queue grows without limit. It takes **4 cash lanes** (75%, wait ≈ 15 s) to recover. Policy and infrastructure trade off directly.

GUIDE 06 TRAFFIC & VEHICLES ADVANCED

Car Service Centre

A workshop receives about 11 cars an hour: advisors take the job, 60% get a quick service and 40% a major repair, then every car is washed.

QUESTION Which bays limit the workshop, and how much demand growth can it absorb?

#	BLOCK	NAME IT	SETTINGS	WHY
1	**Start Point**	**Cars arrive**	**Natural Random Spacing** · Arrival Speed: **0.003**	About 10.8 cars an hour.
2	**Waiting Line → Service Bay (Limited)**	**Reception queue → Service advisors**	Line: defaults · **3** · Time **600** · **Random Service Duration**	Ten minutes to write up a job.
3	**Decision Block**	**Job type**	Paths: **Minor queue 0.6**, **Major queue 0.4**	
4	**Waiting Line → Service Bay (Limited)**	**Minor queue → Quick-service bays**	Line: defaults · **4** · Time **1800** · **Random Service Duration**	30-minute services.
5	**Waiting Line → Service Bay (Limited)**	**Major queue → Repair bays**	Line: defaults · **10** · Time **7200** · **Random Service Duration**	Two-hour repairs.
6	**Waiting Line → Service Bay (Limited)**	**Wash queue → Wash bay**	Line: defaults · **1** · Time **240** · **Random Service Duration**	Both paths merge.
7	**Exit Point**	**Delivered**	Record Final Score: **On**	

```
Cars arrive → Reception queue → Service advisors → Job type
Job type → (0.6) Minor queue → Quick-service bays → Wash queue
Job type → (0.4) Major queue → Repair bays → Wash queue
Wash queue → Wash bay → Delivered
```

RUN Duration: **3 days (two-hour repairs make a one-day run mostly warm-up)**

CHECK THE RESULTS	THEORY	WHY
Cars	≈ 778	0.003 × 259,200
Advisors	60% · wait ≈ 177 s	M/M/3
Quick-service bays	81% · wait ≈ 24 min	M/M/4, λ = 0.0018
Repair bays	86% · wait ≈ 50 min	M/M/10, λ = 0.0012
Wash bay	72% · wait ≈ 10 min	M/M/1

EXPERIMENT

Try: Reliability Lab → Stress: demand **+15%**.

Expect: Repair-bay load rises to ≈ 99%: waits explode. An 11th repair bay brings it back to ≈ 90%. The repair bays, not the advisors, cap growth.

GUIDE 07 LIQUID & MATERIAL EASY

Storm-water Retention Tank

During a storm, inflow pulses reach a retention tank every 10 seconds, each adding 20 units.

QUESTION How long until the tank overflows if nothing is drawn off?

#	BLOCK	NAME IT	SETTINGS	WHY
1	**Start Point**	**Inflow pulses**	**Fixed & Exact Spacing** · Arrival Speed: **0.1**	Exactly one pulse every 10 seconds.
2	**Storage Tank**	**Retention tank**	Capacity **1000** · Initial Level **0** · Fill Rate **20**	In the engine, each arriving pulse adds the Fill Rate amount to the tank.
3	**Exit Point**	**Outfall**	Record Final Score: **On**	Counts pulses the tank accepted.

```
Inflow pulses → Retention tank → Outfall
```

RUN Duration: **1 hr**

CHECK THE RESULTS	THEORY	WHY
Pulses arriving	360	one every 10 s for 3,600 s
Tank full after	50 pulses ≈ 500 s	1,000 ÷ 20
Pulses dropped (overflow)	≈ 310	360 − 50
Tank utilization	100%	level ÷ capacity

EXPERIMENT

Try: Set Capacity to **5000**.

Expect: The tank now lasts 250 pulses, about **42 minutes**, and only ≈ 110 pulses overflow. Sizing question answered: 7,200 units of capacity survives a one-hour storm.

> Tanks fill but are not drained in the current engine; a pulse that does not fit is counted as dropped and never reaches the exit.

GUIDE 08 LIQUID & MATERIAL MEDIUM

Two Tanks, One Valve

A splitter valve sends 70% of incoming batches to Tank A and 30% to Tank B; each tank feeds a filter.

QUESTION Which tank overflows first, and what split avoids overflow for an hour?

#	BLOCK	NAME IT	SETTINGS	WHY
1	**Start Point**	**Inflow**	**Natural Random Spacing** · Arrival Speed: **0.2**	720 batches an hour.
2	**Valve / Splitter**	**Splitter valve**	Paths: **Tank A 0.7**, **Tank B 0.3**	
3	**Storage Tank**	**Tank A**	Capacity **4000** · Initial **0** · Fill Rate **10**	
4	**Storage Tank**	**Tank B**	Capacity **4000** · Initial **0** · Fill Rate **10**	
5	**Processing Unit**	**Filter A**	Average Duration **30** · **Random Service Duration**	Treatment time; many batches can be in the filter at once.
6	**Processing Unit**	**Filter B**	Average Duration **30** · **Random Service Duration**	
7	**Exit Point**	**Treated water**	Record Final Score: **On**	

```
Inflow → Splitter valve
Splitter valve → (0.7) Tank A → Filter A → Treated water
Splitter valve → (0.3) Tank B → Filter B → Treated water
```

RUN Duration: **1 hr**

CHECK THE RESULTS	THEORY	WHY
Batches	≈ 720	0.2 × 3,600
Tank A	full at ≈ 48 min; ≈ 104 dropped	504 batches × 10 = 5,040 > 4,000
Tank B	≈ 54% full	216 × 10 ÷ 4,000
Treated water	≈ 616	400 + 216

EXPERIMENT

Try: Change the split to **0.5 / 0.5**.

Expect: Each tank receives ≈ 3,600 units: both end ≈ 90% full, nothing overflows, and ≈ 720 batches are treated.

GUIDE 09 LIQUID & MATERIAL ADVANCED

Batch Reactor with Quality Split

A plant releases a batch every 5 minutes into a 10-minute reactor cycle; 90% pass QC to the product tank, 10% go to an off-spec tank.

QUESTION How many reactors does the schedule need, and when do the tanks fill?

#	BLOCK	NAME IT	SETTINGS	WHY
1	**Start Point**	**Batch release**	**Fixed & Exact Spacing** · Arrival Speed: **0.003333**	One batch every 300 seconds.
2	**Processing Unit**	**Reactor cycle**	Average Duration **600** · **Fixed Constant Duration**	A Processing Unit has no limit on simultaneous batches, so its average contents tell you how many vessels you need.
3	**Valve / Splitter**	**Quality check**	Paths: **Product tank 0.9**, **Off-spec tank 0.1**	
4	**Storage Tank**	**Product tank**	Capacity **100** · Initial **0** · Fill Rate **1**	Fill Rate 1 counts batches.
5	**Storage Tank**	**Off-spec tank**	Capacity **20** · Initial **0** · Fill Rate **1**	
6	**Exit Point**	**Dispatched**	Record Final Score: **On**	Connect both tanks to it.

```
Batch release → Reactor cycle → Quality check
Quality check → (0.9) Product tank → Dispatched
Quality check → (0.1) Off-spec tank → Dispatched
```

RUN Duration: **8 hrs**

CHECK THE RESULTS	THEORY	WHY
Batches released	≈ 95–96	28,800 ÷ 300
Batches in the reactor at once	2	Little's Law: 600 ÷ 300 → two reactors
Completed by the end	≈ 94	starts after 28,200 s are still cooking
Product tank	≈ 85 batches (85%)	0.9 × 94
Off-spec tank	≈ 9 (≈ 47%)	0.1 × 94

EXPERIMENT

Try: Release a batch every 2 minutes: Arrival Speed **0.008333**.

Expect: Reactors needed = 600 ÷ 120 = **5**. ≈ 212 good batches against a 100-batch tank: the product tank overflows after about **3.9 hours**, so dispatch must run during the shift.

GUIDE 10 MANUFACTURING EASY

Single CNC Machine

One CNC machine receives a blank about every 50 seconds; each part takes 40 seconds.

QUESTION How busy is the machine, and how long do parts wait?

#	BLOCK	NAME IT	SETTINGS	WHY
1	**Start Point**	**Raw blanks**	**Natural Random Spacing** · Arrival Speed: **0.02**	72 blanks an hour.
2	**Waiting Line → Machine / Robot**	**Machine queue → CNC machine**	Line: defaults · Staff/Parallel **1** · Time **40** · **Consistent Around Average**	Machined cycle times vary only a little (the engine uses ±30% spread).
3	**Exit Point**	**Finished parts**	Record Final Score: **On**	

```
Raw blanks → Machine queue → CNC machine → Finished parts
```

RUN Duration: **8 hrs**

CHECK THE RESULTS	THEORY	WHY
Blanks	≈ 576	0.02 × 28,800
Machine utilization	≈ 80%	0.02 × 40
Average wait	≈ 87 s	M/G/1 (Pollaczek–Khinchine)
Parts waiting	≈ 1.7	Little's Law

EXPERIMENT

Try: A worn tool raises the cycle to **45** seconds.

Expect: Utilization 90%, wait ≈ **221 s**: a 12.5% slower cycle makes waiting 2.5× longer.

GUIDE 11 MANUFACTURING MEDIUM

Two-Machine Line with QC Rework

Castings are turned on a lathe, buffered, milled, then checked; 10% fail QC and are reworked before going back to the mill.

QUESTION How much mill capacity does rework consume?

#	BLOCK	NAME IT	SETTINGS	WHY
1	**Start Point**	**Castings**	**Natural Random Spacing** · Arrival Speed: **0.025**	90 an hour.
2	**Waiting Line → Machine / Robot**	**Turning queue → Lathe**	Line: defaults · **1** · Time **30** · **Consistent Around Average**	
3	**Waiting Line → Machine / Robot**	**WIP buffer → Mill**	Line: defaults · **1** · Time **30** · **Consistent Around Average**	Receives from the lathe and from rework.
4	**Decision Block**	**QC check**	Paths: **Finished goods 0.9**, **Rework 0.1**	
5	**Assembly Step**	**Rework**	Average Duration **120** · **Random Service Duration**	Bench rework; several parts at once.
6	**Exit Point**	**Finished goods**	Record Final Score: **On**	

```
Castings → Turning queue → Lathe → WIP buffer → Mill → QC check
QC check → (0.9) Finished goods
QC check → (0.1) Rework → WIP buffer
```

RUN Duration: **8 hrs**

CHECK THE RESULTS	THEORY	WHY
Castings	≈ 720	0.025 × 28,800
Lathe	75% · wait ≈ 49 s	M/G/1
Mill (includes rework)	83% · wait ≈ 82 s	λ = 0.025 ÷ 0.9
Rework passes	≈ 80	0.025 × (0.1 ÷ 0.9) × 28,800

EXPERIMENT

Try: QC pass rate falls to **0.8** (paths 0.8 / 0.2).

Expect: Mill load 94% and the wait ≈ **245 s**, three times longer. Poor quality eats capacity before it shows up as scrap.

GUIDE 12 MANUFACTURING ADVANCED

Welding Bottleneck with Downtime and a Paint Conveyor

Blanks are laser-cut, robot-welded, carried through a paint-curing conveyor oven, inspected, and 5% are scrapped. The welding robot is available 90% of the time.

QUESTION What limits output, and is a second robot or better uptime the better investment?

#	BLOCK	NAME IT	SETTINGS	WHY
1	**Start Point**	**Blanks**	**Natural Random Spacing** · Arrival Speed: **0.02**	72 an hour.
2	**Waiting Line → Machine / Robot**	**Cutting queue → Laser cutters**	Line: defaults · **2** · Time **50** · **Consistent Around Average**	
3	**Waiting Line → Machine / Robot**	**Weld buffer → Welding robot**	Line: defaults · **1** · Time **40** · **Consistent Around Average**	36 s cycle ÷ 0.90 availability = 40 s. Downtime is folded into the cycle.
4	**Assembly Step**	**Paint oven conveyor**	Average Duration **300** · **Fixed Constant Duration**	A conveyor oven: every part spends 5 minutes inside, many at once.
5	**Waiting Line → Machine / Robot**	**Inspection queue → Inspection**	Line: defaults · **1** · Time **30** · **Consistent Around Average**	
6	**Decision Block**	**Final QC**	Paths: **Shipped 0.95**, **Scrapped 0.05**	
7	**Exit Point**	**Shipped**	Record Final Score: **On**	
8	**Exit Point**	**Scrapped**	Record Final Score: **On**	

```
Blanks → Cutting queue → Laser cutters → Weld buffer → Welding robot
Welding robot → Paint oven conveyor → Inspection queue → Inspection → Final QC
Final QC → (0.95) Shipped · (0.05) Scrapped
```

RUN Duration: **8 hrs**

CHECK THE RESULTS	THEORY	WHY
Laser cutters	≈ 50%	0.02 × 50 ÷ 2
Welding robot	80% · wait ≈ 87 s	the bottleneck: highest load
Parts inside the oven	≈ 6	Little's Law: 0.02 × 300
Inspection	60% · wait ≈ 25 s	M/G/1
Shipped / scrapped	≈ 540 / ≈ 28	95% / 5% of ≈ 568 finished; ≈ 8 still in the line

EXPERIMENT

Try: Reliability Lab → Compare: (a) Welding robot staff **2**; (b) availability 98%, Time **36.7**. Then Stress at **+30%** demand.

Expect: (a) weld wait falls to a few seconds; (b) 73.5% load, wait ≈ 55 s. At +30% demand the single robot passes 100% and the line cannot keep up; only (a) holds. Line ceiling today: 1 ÷ 40 s = 90 parts an hour.

Downtime is modelled through the cycle time because the main engine does not yet apply the panel's breakdown (MTBF) fields.

GUIDE 13 LOGISTICS EASY

Conveyor Belt to Packing Station

Parcels ride a 30-metre infeed belt (45 seconds) to a packing station with two packers.

QUESTION Can two packers keep up, and what does the belt add?

#	BLOCK	NAME IT	SETTINGS	WHY
1	**Start Point**	**Parcels**	**Natural Random Spacing** · Arrival Speed: **0.1**	360 an hour.
2	**Conveyor Step**	**Infeed belt**	Average Duration **45** · **Fixed Constant Duration**	A belt moves every parcel at the same speed and carries many at once.
3	**Waiting Line → Loading Dock**	**Packing queue → Packing station**	Line: defaults · **2** · Time **15** · **Random Service Duration**	Loading Dock is the logistics palette's limited block: use it for any station where only N work at once.
4	**Exit Point**	**Packed**	Record Final Score: **On**	

```
Parcels → Infeed belt → Packing queue → Packing station → Packed
```

RUN Duration: **2 hrs**

CHECK THE RESULTS	THEORY	WHY
Parcels	≈ 720	0.1 × 7,200
Parcels on the belt	≈ 4.5	Little's Law: 0.1 × 45
Packer utilization	≈ 75%	0.1 × 15 ÷ 2
Average packing wait	≈ 19 s	M/M/2

EXPERIMENT

Try: Slow the belt to **90** s, then try **1** packer.

Expect: A slower belt doubles the parcels on it (≈ 9) but packing output is unchanged: it adds lead time, not capacity loss. One packer means 150% load and a queue that never clears.

GUIDE 14 LOGISTICS MEDIUM

Parcel Sorter with Three Chutes

A sort hub scans parcels on an induction belt, sorts them to three chutes with their own loading teams, and merges them onto an outbound belt.

QUESTION Is each chute staffed for its share of the volume?

#	BLOCK	NAME IT	SETTINGS	WHY
1	Start Point	Parcels	**Natural Random Spacing** · Arrival Speed: **0.3**	1,080 an hour.
2	Conveyor Step	Induction & scan	Average Duration **20** · **Fixed Constant Duration**	
3	Decision Block	Sorter	Paths: **Chute A 0.5**, **Chute B 0.3**, **Chute C 0.2**	
4	Waiting Line	Chute A / Chute B / Chute C	Three lines, defaults	
5	Loading Dock	Loaders A	**2** · Time **10** · **Random Service Duration**	
6	Loading Dock	Loaders B	**1** · Time **8** · **Random Service Duration**	
7	Loading Dock	Loaders C	**1** · Time **10** · **Random Service Duration**	
8	Conveyor Step	Outbound belt	Average Duration **30** · **Fixed Constant Duration**	All three docks feed it.
9	Exit Point	Shipped	Record Final Score: **On**	

```
Parcels → Induction & scan → Sorter
Sorter → (0.5) Chute A → Loaders A → Outbound belt
Sorter → (0.3) Chute B → Loaders B → Outbound belt
Sorter → (0.2) Chute C → Loaders C → Outbound belt
Outbound belt → Shipped
```

RUN Duration: **1 hr**

CHECK THE RESULTS	THEORY	WHY
Parcels	≈ 1,080	0.3 × 3,600
Loaders A	75% · wait ≈ 13 s	M/M/2, λ = 0.15
Loaders B	72% · wait ≈ 21 s	M/M/1, λ = 0.09
Loaders C	60% · wait ≈ 15 s	M/M/1, λ = 0.06
Parcels on each belt	≈ 6 / ≈ 9	0.3 × 20 and 0.3 × 30

EXPERIMENT

Try: A new customer shifts the sort plan to **0.4 / 0.4 / 0.2**.

Expect: Loaders B jump to 96% and their queue explodes. Moving a loader from A does not work (A would hit 120%); B needs a second loader (48%). Staffing must follow the sort plan.

GUIDE 15 LOGISTICS ADVANCED

Cross-dock: Truck Waves to Outbound Docks

A truck with 120 parcels arrives every 10 minutes; a 3-person team unloads it onto a sortation belt that feeds three outbound docks.

QUESTION Average utilization looks fine. Where do the waves still create queues?

#	BLOCK	NAME IT	SETTINGS	WHY
1	**Start Point**	**Inbound trucks**	**Exact Time Schedule**: Add Arrival six times: at second **0, 600, 1200, 1800, 2400, 3000**, how many **120**	Each entry releases a whole truckload at once.
2	**Waiting Line → Loading Dock**	**Unload queue → Unload team**	Line: defaults · **3** · Time **6** · **Fixed Constant Duration**	Three people, 6 s a parcel: a truck takes 240 s.
3	**Conveyor Step**	**Sortation belt**	Average Duration **60** · **Fixed Constant Duration**	
4	**Decision Block**	**Destination**	Paths: **Dock A line 0.4**, **Dock B line 0.35**, **Dock C line 0.25**	
5	**Waiting Line**	**Dock A / B / C line**	Three lines, defaults	
6	**Loading Dock**	**Dock A**	**2** · Time **12** · **Random Service Duration**	
7	**Loading Dock**	**Dock B**	**1** · Time **10** · **Random Service Duration**	
8	**Loading Dock**	**Dock C**	**1** · Time **10** · **Random Service Duration**	
9	**Exit Point**	**Loaded**	Record Final Score: **On**	All three docks connect here.

```
Inbound trucks → Unload queue → Unload team → Sortation belt → Destination
Destination → (0.4) Dock A line → Dock A → Loaded
Destination → (0.35) Dock B line → Dock B → Loaded
Destination → (0.25) Dock C line → Dock C → Loaded
```

RUN Duration: **1 hr**

CHECK THE RESULTS	THEORY	WHY
Parcels	720	6 trucks × 120
Unload team	40%	720 × 6 ÷ (3 × 3,600)
Dock A / B / C utilization	≈ 48% / 70% / 50%	parcels × time ÷ (docks × 3,600)
Peak queue per wave (A / B / C)	≈ 8 / ≈ 18 / ≈ 6	wave feeds 0.5/s for 240 s vs dock capacity
Loaded	≈ 720	the last wave clears before 3,600 s

EXPERIMENT

Try: Double the unload team to **6**.

Expect: Unloading finishes in 120 s, so parcels hit the docks at 1/s: Dock B's peak queue grows to ≈ 30. Speeding up one step moves the queue downstream; use Reliability Lab → Compare to see it.

> Watch the queue-depth chart rather than averages: the story here is the saw-tooth every 10 minutes.

GUIDE 16 NETWORK & SIGNAL EASY

Fixed Cable Latency

A sender emits a message every 5 seconds over a link with a fixed 10-second delay.

QUESTION What latency and throughput does the receiver see?

#	BLOCK	NAME IT	SETTINGS	WHY
1	**Template**	**Cable / Event Latency**	New **Network & Signal** project → template gallery → **Cable / Event Latency**	The template presets the channel delay, which the panel does not edit today.
2	**Start Point**	**Sender A**	Check: **Fixed & Exact Spacing** · Arrival Speed **0.2**	Editable.
3	**Transport Channel**	**Cable**	Preset: **10 s** fixed delay, unlimited buffer, late detection on	Carries every message with the same delay.
4	**Exit Point**	**Receiver A**	Record Final Score: **On**	

```
Sender A → Cable → Receiver A
```

RUN Duration: **1 hr**

CHECK THE RESULTS	THEORY	WHY
Messages sent	720	0.2 × 3,600
Messages received	≈ 718	the last two are still on the cable
Average latency	10	seconds of simulated time (the chart may label it ms)
Late messages	0	a fixed delay never exceeds itself

EXPERIMENT

Try: Set Sender A to **Natural Random Spacing**.

Expect: The count wanders around 720 but latency stays exactly 10: jitter comes from the link, not the sender.

GUIDE 17 NETWORK & SIGNAL MEDIUM

One Message, Three Subscribers

A publisher sends an event about every 8 seconds; a broadcaster copies it to three subscriber pipes.

QUESTION How much traffic does fan-out create downstream?

#	BLOCK	NAME IT	SETTINGS	WHY
1	**Template**	**Broadcast Pipe (One-to-Many)**	Network & Signal project → template gallery → **Broadcast Pipe (One-to-Many)**	
2	**Start Point**	**Generator A**	Check: **Even Range Spread** · Arrival Speed **0.125**	Gaps between 4 and 12 s, 8 on average.
3	**Broadcaster**	**Broadcast Hub**	No settings	Copies each message to every connected path.
4	**Storage Rack**	**Pipe → A / B / C**	Preset: unlimited	Hold messages until the consumer takes them.
5	**Exit Point**	**Consumer A / B / C**	Record Final Score: **On**	

```
Generator A → Broadcast Hub → Pipe → A / B / C → Consumer A / B / C
```

RUN Duration: **1 hr**

CHECK THE RESULTS	THEORY	WHY
Messages generated	≈ 450	0.125 × 3,600
Each consumer	≈ 450	every copy is delivered
Total completed	≈ 1,350	3× generated: each copy counts
Buffer depth	≈ 0	consumers take messages immediately

EXPERIMENT

Try: Replace the Broadcast Hub with a **Decision Block** (paths 0.34 / 0.33 / 0.33).

Expect: Total completed drops back to ≈ 450: a Decision Block load-balances, a Broadcaster fans out. The same diagram shape, a very different load.

GUIDE 18 NETWORK & SIGNAL ADVANCED

Microservice Gateway Under Load

An API gateway forwards about 5 requests a second through an auth link (random delay, 2 s average, 100 in-flight slots) to fast and slow services.

QUESTION At what request rate does the auth link start dropping traffic?

#	BLOCK	NAME IT	SETTINGS	WHY
1	**Template**	**Microservice Pipeline**	Network & Signal project → template gallery → **Microservice Pipeline**	
2	**Start Point**	**API Gateway**	Check: **Rush Hour / Burst Spikes** · Arrival Speed **5**	Behaves like random arrivals in the engine.
3	**Transport Channel**	**Auth Channel**	Preset: random delay, mean **2 s**, buffer **100**, late detection on	100 messages can be in flight; more are dropped.
4	**Decision Block**	**Service Router**	Preset paths: **Fast 0.4**, **Slow 0.6**	
5	**Storage Rack**	**Fast Queue / Slow Queue**	Preset	
6	**Exit Point**	**Fast / Slow Consumer**	Record Final Score: **On**	

```
API Gateway → Auth Channel → Service Router
Service Router → (0.4) Fast Queue → Fast Consumer
Service Router → (0.6) Slow Queue → Slow Consumer
```

RUN Duration: **10 mins**

CHECK THE RESULTS	THEORY	WHY
Requests	≈ 3,000	5 × 600
Average auth latency	≈ 2	mean of the delay
Late messages	≈ 37%	an exponential delay exceeds its mean e^{-1} of the time
Dropped	≈ 0	≈ 10 in flight vs 100 slots
Fast / Slow	≈ 1,200 / ≈ 1,800	0.4 / 0.6 split

EXPERIMENT

Try: Raise API Gateway Arrival Speed to **40**, **50**, then **60**.

Expect: Drops ≈ **0.4%**, **7.6%**, **19.6%** (Erlang loss with 100 slots and load = rate × 2 s). Late stays ≈ 37%. Here the buffer, not compute, sets the ceiling.

B

Pitch catalogue

Thirty places where discrete-event simulation earns its keep, five per domain.

About these use cases

They are typical industry applications of this kind of simulation and what JustCmul8 can model for each. They are not customer case studies, and no company named or implied here is a JustCmul8 customer.

B PITCH CATALOGUE

People & Service

Palette: Start Point · Waiting Line · Limited Counter · Counter · Decision Block · Exit Point

SIMPLE Bank branch teller staffing

Banking

BUSINESS PROBLEM
Lines at lunch and month-end; customers leave.

WHAT YOU MODEL
Start Point → Waiting Line (patience) → Limited Counter

KPIS → DECISION
Wait time, teller utilization, walk-aways → **Tellers per time band**

SIMPLE Café or quick-service counter

Food service

BUSINESS PROBLEM
Morning rush overwhelms the counter.

WHAT YOU MODEL
Start Point → Waiting Line → Limited Counter → Exit Point

KPIS → DECISION
Wait, queue length, throughput → **Staff on the rush shift**

SIMPLE Metro and railway ticketing

Transit

BUSINESS PROBLEM
Token queues back up into the concourse at peak.

WHAT YOU MODEL
Decision Block for card vs token, two counters in series

KPIS → DECISION
Wait, walk-aways, gate load → **Counters and gates to open**

COMPLEX Hospital OPD and emergency triage

Healthcare

BUSINESS PROBLEM
Long waits, uneven doctor load, pharmacy pile-ups.

WHAT YOU MODEL
Registration → Triage (Decision) → parallel doctor pools → pharmacy

KPIS → DECISION
Wait per stage, utilization, patients leaving → **Doctor rostering, triage rules**

COMPLEX Contact centre with escalation

BPO / support

BUSINESS PROBLEM
Callers abandon; L2 agents idle or swamped.

WHAT YOU MODEL
Waiting Line (patience) → L1 agents → Decision → L2 agents

KPIS → DECISION
Abandon rate, service level, utilization → **Agents per tier and escalation rate**

Traffic & Vehicles

Palette: Start Point · Waiting Line · Service Bay (Limited) · Processing Block · Decision Block · Exit Point

SIMPLE Fuel and CNG station pumps

Retail fuel

BUSINESS PROBLEM
Queues spill onto the road at peak.

WHAT YOU MODEL
Start Point → Waiting Line (patience) → Service Bay

KPIS → DECISION
Wait, drive-aways, pump utilization → **Number of pumps or dispensers**

SIMPLE Toll plaza lane mix

Highways

BUSINESS PROBLEM
Cash lanes back up while tag lanes idle.

WHAT YOU MODEL
Decision Block by payment type → lane pools → Processing Block

KPIS → DECISION
Wait per lane type, utilization → **Lanes per payment type**

COMPLEX EV charging hub sizing

E-mobility

BUSINESS PROBLEM
Chargers either idle or fully booked with long waits.

WHAT YOU MODEL
Decision fast vs slow charge → two Service Bay pools

KPIS → DECISION
Wait, charger utilization, drivers turned away → **Charger count and fast/slow mix**

COMPLEX Car service workshop

Automotive aftersales

BUSINESS PROBLEM
Bays congested; delivery promises missed.

WHAT YOU MODEL
Advisors → Decision (minor/major) → bay pools → wash

KPIS → DECISION
Lead time, bay utilization → **Bays, technicians, booking caps**

COMPLEX Port or terminal truck gate

Ports and logistics parks

BUSINESS PROBLEM
Truck queues outside the gate; demurrage.

WHAT YOU MODEL
Gate lanes → Decision (inspection) → yard Processing Block

KPIS → DECISION
Truck turnaround, queue length → **Gate lanes and inspection rate**

Liquid & Material

Palette: Start Point · Storage Tank · Processing Unit · Valve / Splitter · Exit Point

SIMPLE Storm-water retention sizing

Municipal water

BUSINESS PROBLEM
Will the tank overflow in a design storm?

WHAT YOU MODEL
Start Point (pulses) → Storage Tank → Exit Point

KPIS → DECISION
Time to full, overflowed volume → **Tank size and inflow limits**

SIMPLE Tanker unloading at a depot

Fuel and chemicals

BUSINESS PROBLEM
Receiving tanks fill before the next dispatch.

WHAT YOU MODEL
Exact Time Schedule of tankers → Storage Tank

KPIS → DECISION
Fill time, rejected loads → **Tank capacity and delivery schedule**

SIMPLE Water treatment train

Utilities

BUSINESS PROBLEM
Filters and clear-water tanks mismatched in size.

WHAT YOU MODEL
Valve / Splitter → tanks → Processing Units

KPIS → DECISION
Tank levels, treated volume → **Split ratio and tank sizing**

COMPLEX Dairy or beverage batch processing

Food and beverage

BUSINESS PROBLEM
Pasteuriser and storage tanks limit daily output.

WHAT YOU MODEL
Processing Unit (batch time) → Valve (QA) → product tanks

KPIS → DECISION
Batches in process, tank fill time → **Vessel count and release schedule**

COMPLEX Chemical batch reactor scheduling

Chemicals and pharma

BUSINESS PROBLEM
How many reactors a new schedule needs.

WHAT YOU MODEL
Fixed-spacing releases → Processing Unit → QC split → tanks

KPIS → DECISION
Reactors needed (Little's Law), off-spec volume → **Capex for reactors and tanks**

Manufacturing

Palette: Start Point · Waiting Line · Machine / Robot · Assembly Step · Decision Block · Exit Point

SIMPLE Single machine cell

Discrete manufacturing

BUSINESS PROBLEM
Is the CNC overloaded? How much WIP is normal?

WHAT YOU MODEL
Start Point → Waiting Line → Machine / Robot

KPIS → DECISION
Utilization, wait, WIP → **Shift pattern, second machine**

SIMPLE Paint or curing conveyor oven

Automotive, appliances

BUSINESS PROBLEM
How many parts must the oven hold?

WHAT YOU MODEL
Assembly Step as the oven (fixed time, many at once)

KPIS → DECISION
Parts inside (Little's Law), lead time → **Oven length and conveyor speed**

COMPLEX QC and rework loop

Precision engineering

BUSINESS PROBLEM
Rework silently consumes machine capacity.

WHAT YOU MODEL
Machine → Decision (QC) → Assembly Step (rework) → back

KPIS → DECISION
Rework passes, machine load, wait → **Quality investment vs extra capacity**

COMPLEX Assembly line balancing

Electronics, automotive

BUSINESS PROBLEM
Uneven stations starve or block each other.

WHAT YOU MODEL
Chain of Machine / Robot stations with buffers

KPIS → DECISION
Station utilization, bottleneck, throughput → **Re-balancing tasks between stations**

COMPLEX Bottleneck and downtime (OEE)

Any line with robots

BUSINESS PROBLEM
Second robot or better maintenance?

WHAT YOU MODEL
Availability folded into cycle time; Compare scenarios

KPIS → DECISION
Throughput ceiling, wait, utilization → **Capex vs maintenance programme**

Logistics

Palette: Start Point · Waiting Line · Loading Dock · Conveyor Step · Decision Block · Exit Point

SIMPLE Packing station staffing

E-commerce fulfilment

BUSINESS PROBLEM
Packers cannot keep up with the infeed belt.

WHAT YOU MODEL
Conveyor Step → Waiting Line → Loading Dock

KPIS → DECISION
Packer utilization, wait, belt load → **Packers per shift**

SIMPLE Inbound dock scheduling

Warehousing

BUSINESS PROBLEM
Trucks wait for a free dock.

WHAT YOU MODEL
Exact Time Schedule → Waiting Line → Loading Dock

KPIS → DECISION
Truck wait, dock utilization → **Number of docks and slot booking**

SIMPLE Returns processing

Retail and e-commerce

BUSINESS PROBLEM
Returns pile up; refunds are late.

WHAT YOU MODEL
Inspection → Decision (restock / refurbish / scrap)

KPIS → DECISION
Backlog, lead time → **Inspectors and routing rules**

COMPLEX Parcel sortation hub

Courier and postal

BUSINESS PROBLEM
Chute staffing does not match the sort plan.

WHAT YOU MODEL
Conveyor Steps → Decision (sorter) → chute docks → outbound belt

KPIS → DECISION
Chute wait, loader utilization → **Loaders per chute, sort plan changes**

COMPLEX Cross-dock wave planning

3PL and retail distribution

BUSINESS PROBLEM
Truck waves create temporary queues at outbound docks.

WHAT YOU MODEL
Scheduled truckloads → unload team → belt → docks

KPIS → DECISION
Peak queue per wave, dock utilization → **Unload team size and truck intervals**

B PITCH CATALOGUE

Network & Signal

Palette: Start Point · Transport Channel · Storage Rack · Broadcaster · Decision Block · Exit Point

SIMPLE Link latency budgeting

Telecom and IoT

BUSINESS PROBLEM
Does end-to-end latency meet the SLA?

WHAT YOU MODEL
Start Point → Transport Channel → Exit Point (template)

KPIS → DECISION
Latency, late messages → **Link choice and SLA targets**

SIMPLE Pub/sub fan-out sizing

Event-driven software

BUSINESS PROBLEM
Subscribers multiply downstream traffic.

WHAT YOU MODEL
Start Point → Broadcaster → Storage Racks → consumers

KPIS → DECISION
Messages per consumer, total load → **Consumer capacity and topic design**

COMPLEX API gateway buffer sizing

SaaS and fintech

BUSINESS PROBLEM
Requests dropped at peak.

WHAT YOU MODEL
Gateway → Transport Channel (buffer) → Decision → services

KPIS → DECISION
Drop rate, latency, late % → **Buffer size and rate limits**

COMPLEX IoT telemetry ingestion

Industrial IoT, smart meters

BUSINESS PROBLEM
Burst reporting overwhelms the ingest link.

WHAT YOU MODEL
Burst arrivals → Transport Channel → Storage Rack → processors

KPIS → DECISION
Drops, buffer depth → **Buffer, batching and reporting interval**

COMPLEX Transaction routing (fast and slow paths)

Payments

BUSINESS PROBLEM
Slow path congestion delays settlement.

WHAT YOU MODEL
Decision Block routing → separate processing paths

KPIS → DECISION
Latency per path, throughput → **Routing rules and path capacity**